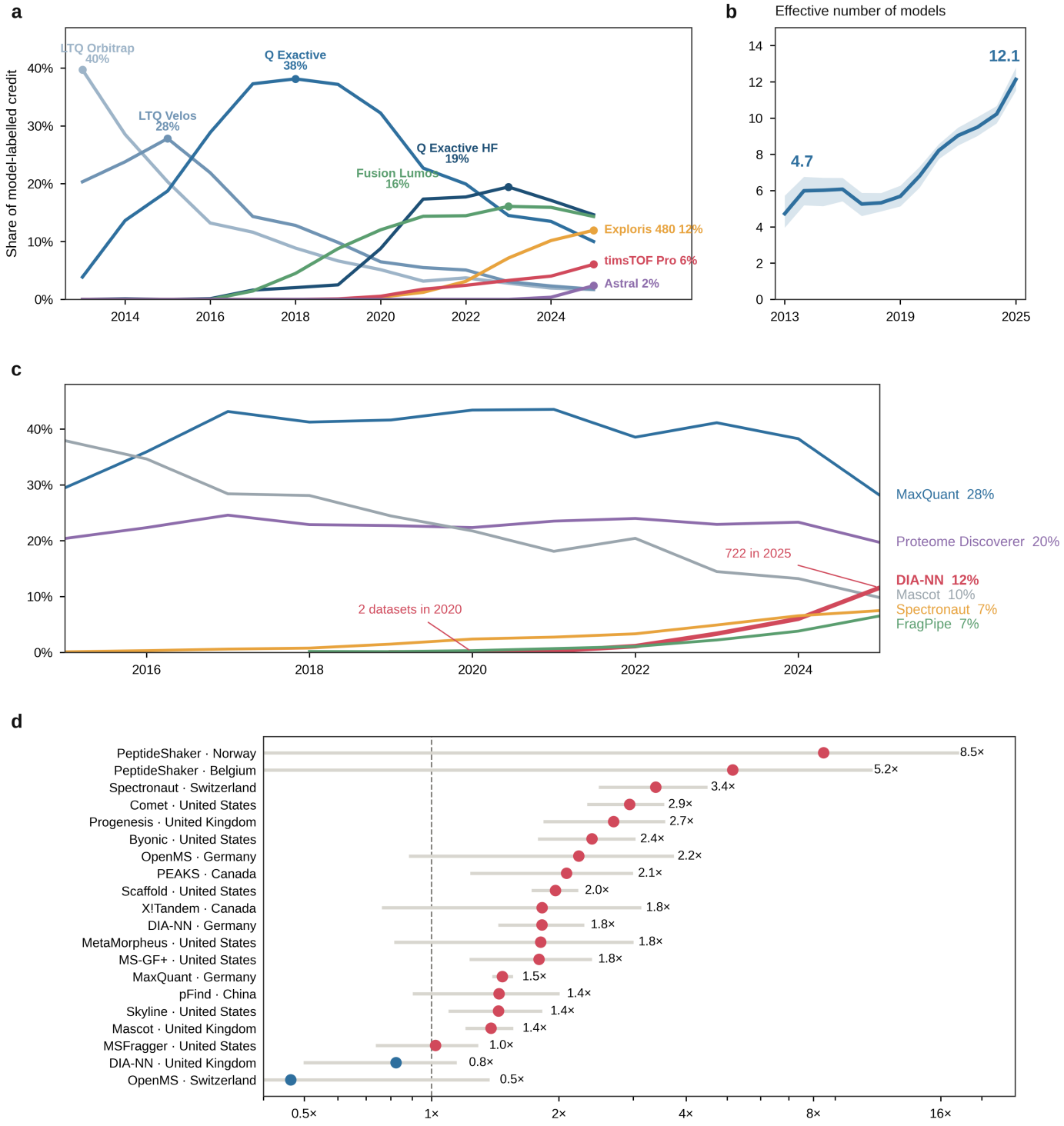

a
Share of model-labelled credit
LTQ Orbitrap
40%
Q Exactive
38%
LTQ Velos
28%
Q Exactive HF
19%
Fusion Lumos
16%
Exploris 480 12%
timsTOF Pro 6%
Astral 2%
0%
10%
20%
30%
40%
2014
2016
2018
2020
2022
2024
b
Effective number of models
12.1
4.7
0
2
4
6
8
10
12
14
2013
2019
2025
c
MaxQuant 28%
Proteome Discoverer 20%
722 in 2025
DIA-NN 12%
Mascot 10%
Spectronaut 7%
FragPipe 7%
2 datasets in 2020
0%
10%
20%
30%
40%
2016
2018
2020
2022
2024
d
PeptideShaker · Norway
PeptideShaker · Belgium
Spectronaut · Switzerland
Comet · United States
Progenesis · United Kingdom
Byonic · United States
OpenMS · Germany
PEAKS · Canada
Scaffold · United States
X!Tandem · Canada
DIA-NN · Germany
MetaMorpheus · United States
MS-GF+ · United States
MaxQuant · Germany
pFind · China
Skyline · United States
Mascot · United Kingdom
MSFragger · United States
DIA-NN · United Kingdom
OpenMS · Switzerland
8.5×
5.2×
3.4×
2.9×
2.7×
2.4×
2.2×
2.1×
2.0×
1.8×
1.8×
1.8×
1.8×
1.5×
1.4×
1.4×
1.4×
1.0×
0.8×
0.5×
0.5×
1×
2×
4×
8×
16×